

Mohammed Abdul Haseeb

mohammed.abd.haseeb@gmail.com | github.com/Abdul-Haseeb-0313

Objective

Aspiring Software Engineer with hands-on experience building backend systems, distributed real-time applications, embedded software, edge AI solutions, and machine learning projects. Enjoy tackling challenging engineering problems and building scalable, efficient software with clean architecture. Seeking a Software Engineering role where I can contribute to impactful products, collaborate with talented engineers, and continuously grow through solving real-world technical challenges.

Education

B.Tech in Computer Science and Engineering	2023–2027
Rajiv Gandhi University of Knowledge Technologies, Basar	CGPA: 8.37
Pre-University Course	2021–2023
Rajiv Gandhi University of Knowledge Technologies	CGPA: 9.56
Secondary Education (SSC)	2020–2021
Silver Wings High School, Nizamabad, Telangana	CGPA: 10.0

Experience

Edge AI Intern (Team Lead)	May 2026 – Jul 2026
Socio Matics Pvt Ltd	

- Led a team of four to develop **Terra-Scope**, a standalone edge-AI soil analyser providing crop and fertiliser recommendations without internet connectivity.
- Architected and implemented the complete software and AI stack, including Modbus communication, robust MAD-based sensor filtering, on-device Decision Tree inference (84% accuracy, <5 ms using emlearn), keypad interface, and voice synthesis.
- Implemented STCR fertiliser equations and built a voice engine using DFPlayer with 222 mapped words for improved accessibility.

Projects

CollabSpace Real-Time Collaborative Whiteboard	Live Frontend Backend
---	---------------------------

- Built a collaborative workspace supporting real-time drawing, role-based permissions, and member management.
- Designed a delta-sync protocol reducing network payload by nearly 99% while combining in-memory caching with PostgreSQL auto-save for fast loading.
- **Tech:** React, Node.js, Socket.IO, PostgreSQL, Tailwind CSS, Excalidraw, Vercel, Render.

ChatApp Real-Time Messaging	Live Frontend Backend
--------------------------------------	---------------------------

- Developed a real-time messaging platform with typing indicators, online presence, and unread message tracking.
- Implemented JWT authentication with token versioning for secure server-side logout and solved message duplication using a socket-driven synchronization model.
- **Tech:** React, Express, Socket.IO, PostgreSQL, Prisma, JWT, Tailwind CSS.

DeepC Deep Learning Library in C	GitHub
---	--------

- Built a neural network library from scratch in C99 featuring dense layers, activation functions, SGD, Adam, Xavier initialization, and model serialization.

- Implemented the complete training pipeline with batch gradient descent, preprocessing utilities, and multiple loss functions.
- **Tech:** C99, GCC, Make, Linear Algebra.

On-Device Audio Anomaly Detector

GitHub

- Developed a convolutional autoencoder on mel-spectrograms for privacy-preserving anomaly detection.
- Quantized the model to 0.33 MB TFLite with 14.6 ms inference on CPU.
- **Tech:** Python, TensorFlow/Keras, TFLite, Librosa, scikit-learn.

Boids Flocking Simulation

Live | GitHub

- Built an interactive flocking simulation featuring predator AI and steering behaviours including alignment, cohesion, and separation.
- Optimized rendering for 100+ agents while demonstrating vector mathematics, OOP, and event-driven programming.
- **Tech:** HTML5 Canvas, JavaScript, CSS.

Truth Table Printer | Logical Expression Evaluator

GitHub

- Developed a Python library that generates truth tables for arbitrary logical expressions using the Shunting Yard algorithm for infix-to-postfix conversion.
- Supports operators: NOT, AND, OR, Implication, and Biconditional; prints formatted truth tables directly from a simple function call.
- **Tech:** Python, Shunting Yard Algorithm, CLI.

Skills

Languages: C, C++, Java, Go, Python, JavaScript

Backend: Node.js, Express, REST APIs, Socket.IO, Prisma

Databases: PostgreSQL, MySQL

Frontend: React, Tailwind CSS, HTML5, Canvas API

Machine Learning: TensorFlow/Keras, TFLite, scikit-learn, emlearn

Embedded Systems: ESP32, Modbus, UART, I2C

Tools: Git, Linux, Docker, CMake, Render, Vercel, Maven

Core CS: Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks

Achievements

- Secured **First Place** in two university-level coding competitions.
- Secured **12th Place** in a multi-university coding contest organised by **9am** across three institutions including RGUKT Basar.